Example 8.18 In a distribution, the mean and mode are 66 and 60 respectively. Calculate the median.

Solution

Given, Mean = 66 and Mode = 60.

Using, $\text{Mode} \simeq 3\text{Median} - 2\text{Mean}$

$60 \simeq 3\text{Median} - 2(66)$

$3\ \text{Median} \simeq 60 + 132$

Therefore, $\text{Median} \simeq \frac{192}{3} \simeq 64$

Exercise 8.3

1. The monthly salary of 10 employees in a factory are given below :
   ₹5000, ₹7000, ₹5000, ₹7000, ₹8000, ₹7000, ₹7000, ₹8000, ₹7000, ₹5000
   Find the mean, median and mode.

2. Find the mode of the given data : 3.1, 3.2, 3.3, 2.1, 1.3, 3.3, 3.1

3. For the data 11, 15, 17, x+1, 19, x–2, 3 if the mean is 14 , find the value of x. Also find the mode of the data.

4. The demand of track suit of different sizes as obtained by a survey is given below:

Size	38	39	40	41	42	43	44	45
No. of Persons	36	15	37	13	26	8	6	2

   Which size is in greater demanded?

5. Find the mode of the following data:

Marks	0-10	10-20	20-30	30-40	40-50
Number of students	22	38	46	34	20

6. Find the mode of the following distribution:

Weight(in kgs)	25-34	35-44	45-54	55-64	65-74	75-84
Number of students	4	8	10	14	8	6

8-Stats.indd 287 02/12/2022 21:51:36